Yash Rajesh Garad

Final Year Undergraduate, B.E. Artificial Intelligence | St. Vincent Pallotti College of Engineering & Technology, Nagpur

yash.garad27@gmail.com +91-9370471759 github.com/Garad9901 linkedin.com/in/yash-garad-23a315255

ORCID 0009-0007-3929-9248 tensor-os-garad9901.vercel.app

Education

B.E. Artificial Intelligence *2023 – 2027*

St. Vincent Pallotti College of Engg. & Tech., Nagpur **CGPA 8.32/10**

Experience

RNR Innotech *Jun'26 – Present*

AI Developer Intern

- Building AI modules for **RIM ai ERP**: **invoice scanning**, **inventory forecasting**, a 24/7 **business chat assistant** and **analytics reports**.

Acube AI *Jun'26 – Present*

ML Engineer Intern

- Machine learning engineering on applied AI products.

MRSAC, Nagpur *1 Aug'26 – 1 Sep'26*

Research Intern

- Built a seven-layer **pluvial-flood nowcasting digital twin** for **Kolhapur city** on open remote-sensing data and **back-tested** its skill honestly.

AI Developer Lead, SVPCET *2024 – Present*

- Leads student AI builds in the **Department of Artificial Intelligence**.

Publications (14)

- **Presented:** *Skill Sphere: AI-Driven Personalized Skill Assessment and Career-Oriented Learning*, ICAPSAS-26, Trivandrum, Aug 2026.
- **Under revision:** *FireVisionNet: Hybrid Deep Learning for Forest Fire Detection*, Kufa J. of Engineering.
- **Submitted:** *AI Predictive Analytics and Supply Chain Resilience in Indian Retail*, IJMDM; *PPE Compliance Monitoring with YOLOv11*, ICSIT 2026.
- **In preparation:** Agentic linear algebra tutor (corresponding author); offline RAG–SQL for university data; CLARITY-CT (XAI for chest CT); Kolhapur flood digital twin; physics-assisted renewable forecasting; agentic RAG + Text-to-SQL review; V2V collision risk over LoRa; EEG depth of anaesthesia; agentic AI for SME ERP; BioShield-Rail.

Achievements

- **1st rank**, Microsoft Nirmaan AI Hackathon.
- **3rd Prize**, Webbotix National Departmental Level Hackathon, Dept. of AI, SVPCET.
- **2nd best project** (software category) for College Assistant.
- **Top 15** in multiple national-level hackathons and innovation competitions.

Technical Skills

- **Programming:** Python, C, SQL, R, TypeScript, JavaScript
- **Analytics:** Power BI, Tableau, Excel, Statistical Analysis, EDA
- **AI/ML:** ML, Deep Learning, Gen AI, RAG, LangChain, LangGraph, MCP, GNNs, Transformers, YOLO, OpenCV
- **Frameworks:** FastAPI, React, Next.js, NestJS, Django, TensorFlow, PyTorch, Scikit-Learn, Pandas
- **Databases:** MySQL, PostgreSQL, MongoDB, Firebase, PostGIS, Qdrant
- **Cloud & DevOps:** Docker, Kubernetes, OCI, AWS, GitHub Actions, MLflow, Airflow, Kafka, Terraform
- **Soft Skills:** Problem Solving, Critical Thinking, Adaptability, Teamwork, Communication

Relevant Courses

Data Structures & Algorithms • Machine Learning • Deep Learning • Natural Language Processing • Computer Vision • Generative AI • Cloud Computing • Database Systems

Key Projects

Skill Sphere *Aug'26*

- AI mobile assessment platform (**presented at ICAPSAS-26**) with **real-time adaptive testing** and **GenAI question generation** over a hybrid AI + expert-curated question bank.
- App-level **integrity layer**: screenshot restriction, disabled backtracking, **behavioural anomaly detection**, violation escalation and auto-submit; **email OTP** and token sessions.

College Assistant *Aug'26 – Oct'26*

- Major project, **2nd best (software)**: fully **offline multi-agent assistant** (router, SQL, RAG, synthesis, **verification** agents) over university data on **Ollama**, **Qdrant**, **PostgreSQL**, Django + React and Docker Compose.

Agentic Linear Algebra Tutor *Oct'26 – Present*

- Multi-agent tutor whose **SymPy verification agent** checks every claim and student step; **misconception taxonomy** from real solutions, benchmarked vs. unverified LLMs.

EdgeSeiz-Net *Sep'26 – Oct'26*

- Lightweight **EEG seizure detector** for **ESP32**, trained **patient-independently** on **CHB-MIT** and evaluated on 159 h of unseen-patient EEG.

AgroSense AI *May'26*

- **Rainfall**, **crop-yield** and **climate-risk** forecasting for 11 Vidarbha districts with **XGBoost, LSTM, Transformers, GNNs**, FastAPI, MLflow, PostgreSQL and farmer alerts.

AadhaarX *May'26 – Jun'26*

- **ArcFace face matching**, **liveness detection**, fraud intelligence and audit trail on **Next.js, NestJS, TypeScript**, Docker and Kubernetes.

RouteIQ X *Jul'26*

- Geospatial AI for **predictive road maintenance** and infrastructure health with FastAPI, React, PostGIS, Kafka, Airflow, XGBoost and LSTM.

More Projects

- **IntruShield** – NSL-KDD intrusion detection with SHAP.
- **PPE Compliance Vision** – YOLO workplace-safety monitoring.
- **Multilingual Sarcasm Detection** – code-mixed text, BiLSTM vs. XLM-R.
- **CLARITY-CT Evidence** – evidence base for an XAI review.
- **MathGPT** – Gemini + Lean 4 + SymPy maths.
- **ADHYAYAN** – offline study-assistant agent.
- **SwarKavach** – voice-clone detection, SIH 2026.
- **PolicyPulse** – MGNREGA analytics and NLP.
- **UrduGuard** – Urdu moderation with LIME.
- **Accident Detection System** – real-time YOLO crash alerts.
- **MedBridge AI** – rural triage (MERN, Gemini).
- **GAIA Geospatial Dashboard** – census, NDVI and seismic GIS.
- **Smart Resume Screener** – TF-IDF resume–job matching.
- **AFTERMIND** – LLM cognitive twin graph.
- **LoRa V2V ADAS** – V2V collision warnings.
- **Food Waste Reduction System** – donor–NGO matching.

Certifications

OCI 2025 Certified Generative AI Professional (Oracle) • Fundamentals of Deep Learning (NVIDIA DLI) • Introduction to Machine Learning (NPTEL) • Geocomputation and Geo-web Services (IIRS, ISRO) • Introduction to LangGraph, LangSmith Essentials, Agent Observability & Evaluations (LangChain Academy) • Claude with the Anthropic API, Introduction to MCP (Anthropic) • Career Essentials in Generative AI (Microsoft & LinkedIn) • Foundations of Prompt Engineering (AWS) • AI Security & Governance (Securiti) • Introduction to Modern AI (Cisco) • AI Appreciate, AI for Entrepreneurship (Intel) • ML Dimensionality Reduction (IBM) • Chatbot Development (Nasscom Foundation) • AI in Pharma (NCVET, Govt. of India).